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Physics Informed Neural Network (PINN) for Noise Suppression in Distributed Acoustic Sensing for Vertical Seismic Profiling and CO_2 Storage Monitoring

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Abstract

Distributed Acoustic Sensing (DAS) has emerged as a transformative technology for large-scale seismic and CO_2 storage monitoring, but field data are frequently contaminated by complex, nonstationary noise that obscures weak subsurface signals. Conventional Machine learning/Deep learning based denoising approaches including convolutional neural networks, autoencoders, and transform-domain filters often require large paired noisy–clean datasets and risk overfitting to empirical patterns, making them unreliable in ground-truth-free deployments. Physics-Informed Neural Networks (PINNs) overcome this limitation by embedding the governing partial differential equations directly into the learning process, enabling noise suppression that remains consistent with physical wave propagation. In this work, we develop and optimize PINN architecture for DAS denoising without any clean dataset using the damped acoustic wave equation as a physics prior. Through staged tuning of network depth, neuron count, learning rate, collocation density, and supervised-to-physics loss weighting, the best configuration achieved a 97% increase in low-frequency energy concentration, 39% reduction in spectral entropy, 43% narrowing of −3 dB bandwidth, and 55% gains in both spatial and temporal smoothness, while preserving overall signal energy. The results demonstrate the potential of PINNs as a robust, physically grounded alternative to purely data-driven DAS denoising methods.

Introduction

Distributed Acoustic Sensing (DAS) has revolutionized the way subsurface environments are monitored, enabling high-resolution, real-time acoustic measurements by converting standard optical fibers into dense seismic arrays[1]. Its applications span seismic imaging, infrastructure monitoring, and CO_2 storage surveillance. However, DAS signals are frequently plagued by high levels of coherent and incoherent noise that obscure critical wavefronts, especially in low-amplitude or low-SNR scenarios[2].

Traditional denoising techniques such as Fourier filtering, wavelet thresholding, or singular spectrum analysis offer partial improvements but are inherently limited. They often rely on manually tuned parameters, make strong assumptions about signal stationarity, or risk attenuating important waveform features along with the noise. Moreover, these methods are typically decoupled from the underlying physics of wave propagation, resulting in suboptimal generalization across varying acquisition conditions[3].

Recent advances in machine learning have introduced purely data-driven denoising models, but their applicability to DAS remains constrained by their dependence on large, labeled training sets and their lack of physical interpretability. These black-box models can recover signal-like patterns but may hallucinate features, fail to generalize, or violate the physics governing DAS propagation. In recent years, several advanced machine learning (ML) strategies have been proposed for denoising DAS data. For instance, DAS-N2N adopts a weakly supervised Noise2Noise framework, training on paired noisy recordings from spliced fiber segments to suppress incoherent noise without requiring clean labels yet it often struggles to generalize across varying noise regimes or to enforce physical consistency in under-sampled domains[4] Another promising approach employs a GC-AB-Unet utilizing self-attention and wavelet guidance to denoise DAS traces without labels; however, its reliance on patch-based reconstruction can introduce inconsistent smoothing artifacts, especially in low-SNR events[5] More recently, diffusion models have been explored for DAS denoising; while they effectively recover signals with minimal leakage, their reliance on extensive synthetic training sets introduces challenges in adapting to field variability. Truly unsupervised strategies like masked auto-encoders (MAEs) show promise in handling noisy DAS data, but they often lack the ability to enforce underlying physical wave propagation constraints[6].

Complementing this, a guided unsupervised deep learning network integrates a band-pass filter and continuous wavelet transform (CWT) features into a DL model with self-attention to attenuate noise while reconstructing coherent wave patterns, yet the reliance on pre-filtering may discard subtle signal elements[7]. Another unsupervised architecture employs dense encoding-decoding with residual and skip connections to untrained field DAS data, achieving effective noise suppression, but it can sometimes misconstrue incoherent noise as signal when features overlap[8]. Moreover, spatial deep-deconvolution U-Nets tailored for traffic-DAS can extract vehicle-induced wavelets and resolve overlapping sources, though these methods are tuned to specific band-limited vehicular signals and struggle in general seismic contexts[9].

Across these models, a consistent limitation is their dependence on data-driven patterns alone without embedding the physics of wave propagation, such methods may eliminate noise but risk erasing subtle coherent features or generating physically inconsistent artifacts. To address these shortcomings, we propose adopting a Physics-Informed Neural Network (PINN) framework, embedding the damped 1D wave equation into the learning process. By balancing supervised denoising with PDE residual minimization, PINN enables coherent reconstruction, noise suppression, and physical plausibility in scenarios where clean labels are unavailable a crucial advantage for reliable DAS data interpretation in field-scale applications[10–12].

In this study, we design and systematically optimize a PINN architecture to suppress noise in a field DAS dataset without access to clean ground-truth signals. The model embeds the damped wave equation into its loss formulation, enabling it to enhance signal-to-noise ratio, attenuate residual noise, and reconstruct coherent waveform structures under realistic, ground-truth-free conditions. Beyond conventional error metrics, we evaluate performance using a suite of physics- and signal-based parameters including frequency-band strength, spectral entropy, bandwidth reduction, spatial/temporal smoothness, adjacent-channel coherence, residual noise power, and signal-energy preservation providing a multi-faceted assessment that reflects practical constraints and performance priorities in real-world DAS monitoring scenarios.

Methodology

Field DAS Dataset and Noise Characteristics

The dataset used in this study originates from a publicly available field-acquired Vertical Seismic Profiling (VSP) DAS dataset, obtained from the PubDAS repository [13]. The acquisition represents a real borehole deployment in an operational field setting, with DAS channels corresponding to spatially distributed fiber segments along the wellbore and temporal samples capturing the seismic wavefield over time. The raw data contains naturally occurring noise from field operations, including high-frequency jitters, amplitude spikes, and reduced coherence between adjacent channels characteristics typical of DAS recordings affected by environmental disturbances, surface activity, and acquisition system limitations.

Governing Equation and Physics-Informed Formulation

The backbone of the PINN framework in this study is the incorporation of the 1D damped wave equation, which governs the propagation of seismic strain or particle velocity signals in the DAS system along the vertical axis of the borehole as shown in eq. (1)

$$\frac{\partial^2 u(x,t)}{\partial t^2} + \alpha \frac{\partial u(x,t)}{\partial t} = c^2 \frac{\partial^2 u(x,t)}{\partial x^2} \qquad \text{(eq. 1)}$$

Where:

u(x,t) is the DAS-measured strain or displacement field,

$x \in [0,1]$ is the normalized spatial coordinate along the borehole depth,

$t \in [0,1]$ is the normalized time domain,

c is the wave propagation speed (assumed constant),

α is the damping coefficient accounting for energy loss due to intrinsic attenuation, friction, and imperfect coupling in the DAS system.

This damped wave equation better represents real DAS signals compared to the ideal (undamped) wave equation, especially in field-acquired data such as VSP, where signals experience rapid attenuation with depth and time. The damping term α(∂u/∂t) plays a critical role in suppressing non-physical high-frequency oscillations and improving the model's robustness in noisy regimes.

To enforce this physics in the network, we define the PDE residual term as shown in eq. (2)

$$f_{PDE}(x,t) = \frac{\partial^2 \hat{u}}{\partial t^2} + \alpha \frac{\partial \hat{u}}{\partial t} - c^2 \frac{\partial^2 \hat{u}}{\partial x^2} \qquad \text{(eq. 2)}$$

Where:

u(x,t) is the predicted output of the PINN. This residual is evaluated at randomly sampled collocation points across the spatiotemporal domain and forms the basis of the physics loss term. The corresponding PDE loss function is defined by eq. (3)

$$\mathcal{L}_{PDE} = \frac{1}{N_c} \sum_{i=1}^{N_c} \left(f_{PDE}(x_i, t_i) \right)^2 \qquad \text{(eq. 3)}$$

Where:

N_c is the number of collocation points. This term penalizes deviations from the wave physics and is included in the overall loss function of the network as shown in eq. (4)

$$\mathcal{L}_{total} = \lambda_{sup} \cdot \mathcal{L}_{sup} + \lambda_{PDE} \cdot \mathcal{L}_{PDE} + \lambda_{BC} \cdot \mathcal{L}_{BC} \qquad \text{(eq. 4)}$$

Here, L_{sup} is the supervised mean squared error loss between the PINN output and the noisy DAS data at observed points, L_{BC} enforces Dirichlet boundary conditions, and the λ coefficients are tuneable weights to balance the influence of each term. The supervised loss is given by eq. (5).

$$\mathcal{L}_{sup} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{u}(x_i, t_i) - u_{noisy}(x_i, t_i)\right)^2 \quad \text{(eq. 5)}$$

Training the network to minimize L_{total} enables the PINN to produce outputs that are both data-consistent and physically meaningful. Preliminary experiments showed that excluding the damping term (i.e., using the undamped wave equation) resulted in poor generalization and overfitting to noise. In contrast, the inclusion of damping improved convergence stability and produced smoother, more interpretable signals making it a critical component of the governing equation in this PINN denoising framework.

Hyperparameter Tuning and Optimization

Hyperparameter tuning was conducted in a staged manner, progressively refining the PINN configuration to achieve maximum denoising performance. The learning rate was tested at 1×10^{-3} and 1×10^{-4}, with training epochs ranging from 2,000 to 15,000 depending on convergence behaviour. The supervised loss weight (λ_{sup}) was varied between 5 and 20, while the PDE loss weight (λ_{PDE}) was adjusted from 1 up to 100 to explore physics-dominant regimes. The optimizer was switched from Adam to Adamax. Network width was increased from 50 to 100 neurons per layer in some configurations, and depth varied from 3 to 7 hidden layers. The number of collocation points (*Nc*) was progressively increased from 5,000 to 20,000 to improve PDE constraint enforcement. In one configuration, Savitzky–Golay filtering was applied as a post-processing step to further enhance smoothness. The best-performing model balanced physics and data terms with a compact 3×50 architecture using Swish activation and Adam optimization.

Evaluation Matrices

To assess the performance of the proposed PINN in denoising DAS signals, we report a set of complementary metrics designed to capture improvements in the spectral, spatio-temporal, and physics-consistency domains. In the frequency domain, we evaluate the frequency-band strength ($E_{[0,ft]}/E_{total}$) to quantify the proportion of energy concentrated in the physically plausible 0–0.20 Hz band, spectral entropy ($H= -\sum p(f)\log p(f)$) to measure spectral randomness, and the −3 dB bandwidth to assess the suppression of high-frequency noise. Spatial and temporal coherence are examined using the smoothness in space ($\mathrm{Var}(\partial u/\partial x)$ and smoothness in time ($\mathrm{Var}(\partial u/\partial t)$ both of which should decrease as the signal becomes more coherent. We also calculate adjacent-channel coherence, defined as the mean Pearson correlation between consecutive channels, to capture restored continuity in spatial wave propagation. Noise suppression effectiveness is quantified through residual noise power, computed as the variance between the noisy and denoised signals, while signal-energy preservation, given by the ratio $\mathrm{Var}(u_{PINN})/\mathrm{Var}(u_{noisy})$, ensures that denoising does not excessively attenuate the signal. Finally, physics-consistency is measured via the PDE residual norm, representing the mean squared residual of the damped wave equation over all collocation points. All metrics are computed without access to ground-truth clean data, reflecting realistic DAS deployments, and improvements are reported as relative changes from the noisy baseline.

PINN Architecture

The PINN is implemented as a fully connected feed-forward neural network with the following architecture:

Input Layer: 2 neurons (space *x*, time *t*)
Hidden Layers: 3 layers, each with 50 neurons
Activation Function: Swish (nonlinear, smooth, and differentiable)
Output Layer: 1 neuron (predicted signal u)
Weight Initialization: Xavier normal

The model receives collocation points across the spatiotemporal domain and predicts the signal value *u(x,t)* for each point a shown in Figure 1.

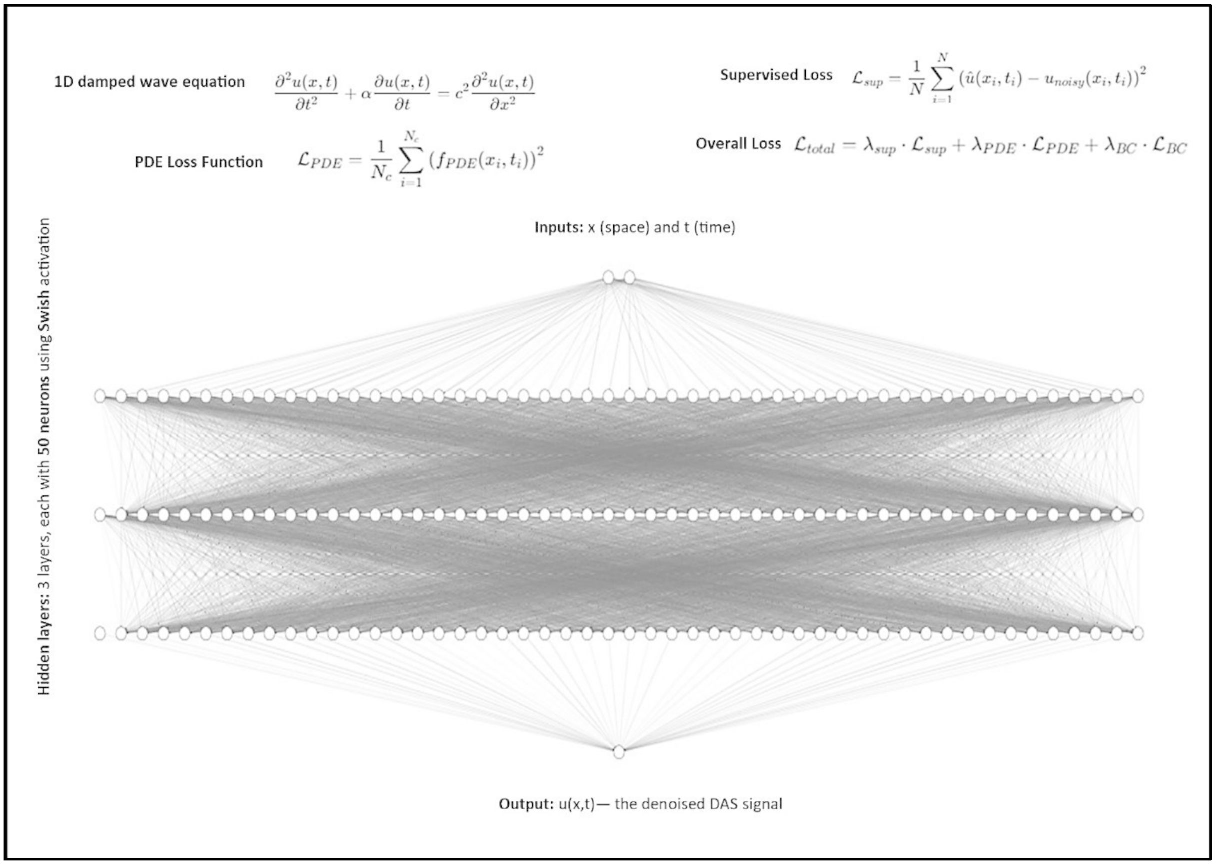

Figure 1—Physics-Informed Neural Network (PINN) architecture used in this study, consisting of three hidden layers with 50 neurons each and Swish activation. Inputs are spatial position x and time t; the output is the denoised DAS signal $u(x,t)$. The model jointly minimizes a supervised loss term and a PDE-based loss term derived from the 1D damped wave equation, with the total loss being a weighted sum of supervised, PDE, and boundary condition losses.

Results and Discussion

Frequency-Domain Analysis Across Multiple Channels

To further evaluate the performance of the proposed PINN framework, frequency–time spectrograms were generated for multiple DAS channels, enabling a channel-wise comparison between noisy and denoised data. Figure 2(a) illustrates the noisy signal spectrograms for Channels 0, 1, 2, 3, 4, and 5, while Figure 2(b) depicts the corresponding outputs after denoising via the PINN model.

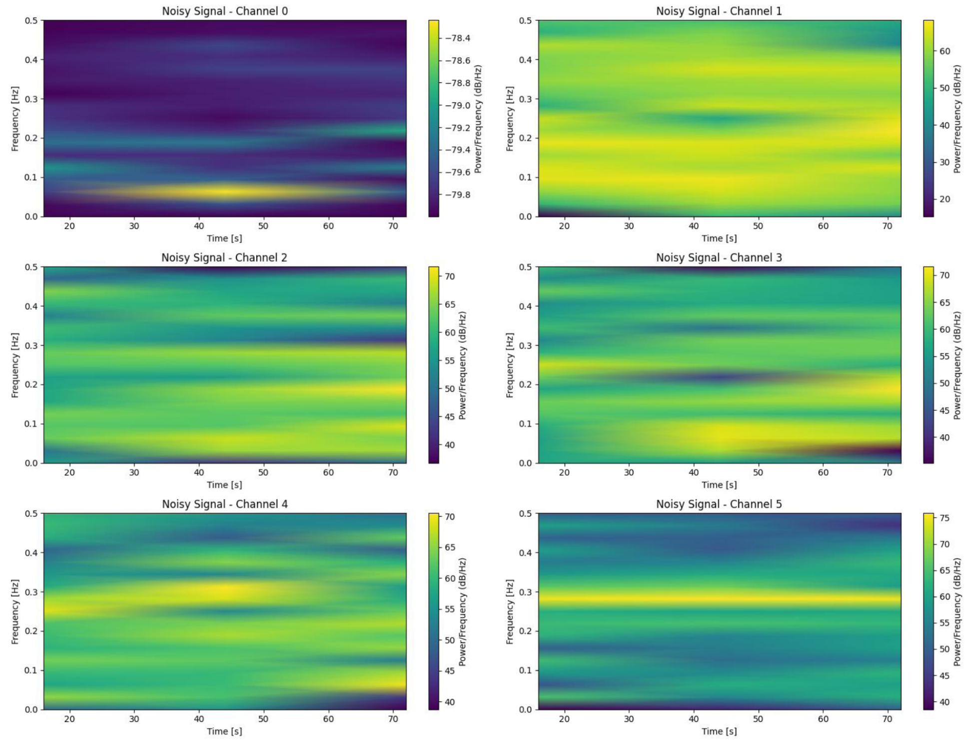

Figure 2(a)—Time–frequency spectrograms of six representative channels from the noisy DAS dataset. The colour scale represents power spectral density (dB/Hz), showing the dispersion of energy across both low- and high-frequency bands due to noise contamination.

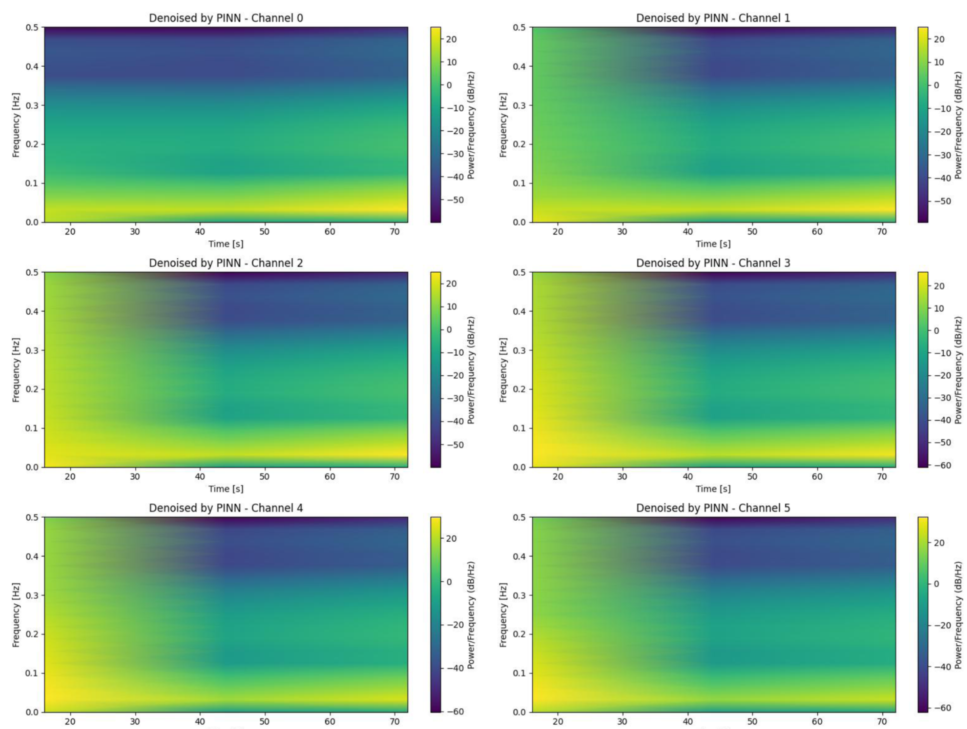

Figure 2(b)—Time–frequency spectrograms of denoised DAS signals across six representative channels (0–5) using the optimized PINN configuration. The results show a clear concentration of signal energy in the physically plausible low-frequency band (<0.2 Hz) and suppression of broadband high-frequency noise, while maintaining consistent spectral structure across channels.

Characteristics of Noisy Channel Data. In the noisy spectrograms (Figure 2a), the power distribution is dispersed across a broad frequency band, with multiple spurious peaks and inconsistent temporal continuity. High-frequency noise components are prominent, masking the underlying low-frequency structures. Notably, the power–frequency scale for noisy channels spans up to 70–75 dB/Hz, suggesting an inflated amplitude profile due to noise. This broadband contamination obscures the temporal coherence of the primary signal, making it challenging to extract event-related patterns.

PINN Denoising Impact on Frequency Profiles. Post-denoising (Figure 2b), the spectrograms reveal a substantial reduction in high-frequency noise components, resulting in a more compact and physically consistent frequency signature. The dominant energy is now localized in the 0–0.15 Hz range across all channels, consistent with the expected low-frequency nature of the underlying seismic response. The dynamic range has shifted, with peak power levels reduced to 20–25 dB/Hz, reflecting both noise attenuation and amplitude normalization inherent to the PINN solution. Temporal continuity is markedly improved, with smoother frequency evolution and the absence of random power spikes.

Cross-Channel Consistency and Physical Plausibility. An important observation is the enhanced cross-channel coherence post-denoising, where frequency–time patterns align more consistently across Channels 0–5. In contrast, the noisy data show channel-to-channel variability that is unlikely to arise from physical propagation effects alone, indicating stochastic noise dominance. The PINN's ability to enforce underlying PDE constraints results in spectrograms that are not only cleaner but also physically plausible, as reflected by the suppression of isolated high-power artifacts and preservation of dominant low-frequency trends.

Critical Perspective. While the denoised spectrograms demonstrate significant noise suppression, a degree of over-smoothing is observed particularly in higher-frequency zones. This effect is not due to an over-dominant physics loss (because physics loss wasn't dominant), but more likely stems from the architectural simplicity of the 1D PINN formulation, which may struggle to represent sharper signal gradients or local spectral fluctuations. Future work involving 2D spatiotemporal PINNs or frequency-aware loss formulations could improve resolution without sacrificing denoising performance.

Denoising Performance Assessment

The performance of the optimized PINN configuration is illustrated through these complementary perspectives: spatiotemporal amplitude maps, mid-time spatial signal comparison, and frequency-domain analysis (Figures 3a. 3b).

Figure 3(a) presents the raw noisy DAS dataset (top) and the corresponding PINN-denoised output (bottom). The original data exhibits high spatial and temporal variability with no discernible coherent wavefronts, characteristic of heavy broadband noise contamination. In contrast, the PINN output reveals a smooth, continuous spatiotemporal profile, representing the underlying wave propagation dynamics as constrained by the governing PDE. While the output appears highly smoothed, this behaviour reflects the physics-governed regularization inherent in the PINN framework, which suppresses high-frequency noise and enforces physical plausibility even in the absence of ground-truth clean data. The reduced dynamic range (colourbar scaling from ~0.1 to ~0.55) indicates amplitude stabilization across the domain. This trade-off between high-frequency retention and noise suppression is a known feature of PINN-driven denoising when the supervised term is weighted to prioritize PDE consistency over raw amplitude preservation.

Figure 3(b) shows the frequency-domain representation of both the noisy input (left) and PINN output (right). The noisy spectrum is dominated by dispersed high-frequency components, with energy spread across the full frequency range, indicating stochastic contamination. The PINN output spectrum is strongly localized along the low-frequency vertical and horizontal axes, with negligible power in the high-frequency bands. This transformation reflects the PINN's suppression of incoherent frequency content while retaining the principal low-frequency components aligned with the underlying PDE solution space. Such spectral concentration is an important metric when SNR cannot be computed due to the absence of a ground-truth signal, as it quantitatively demonstrates the elimination of broadband noise energy.

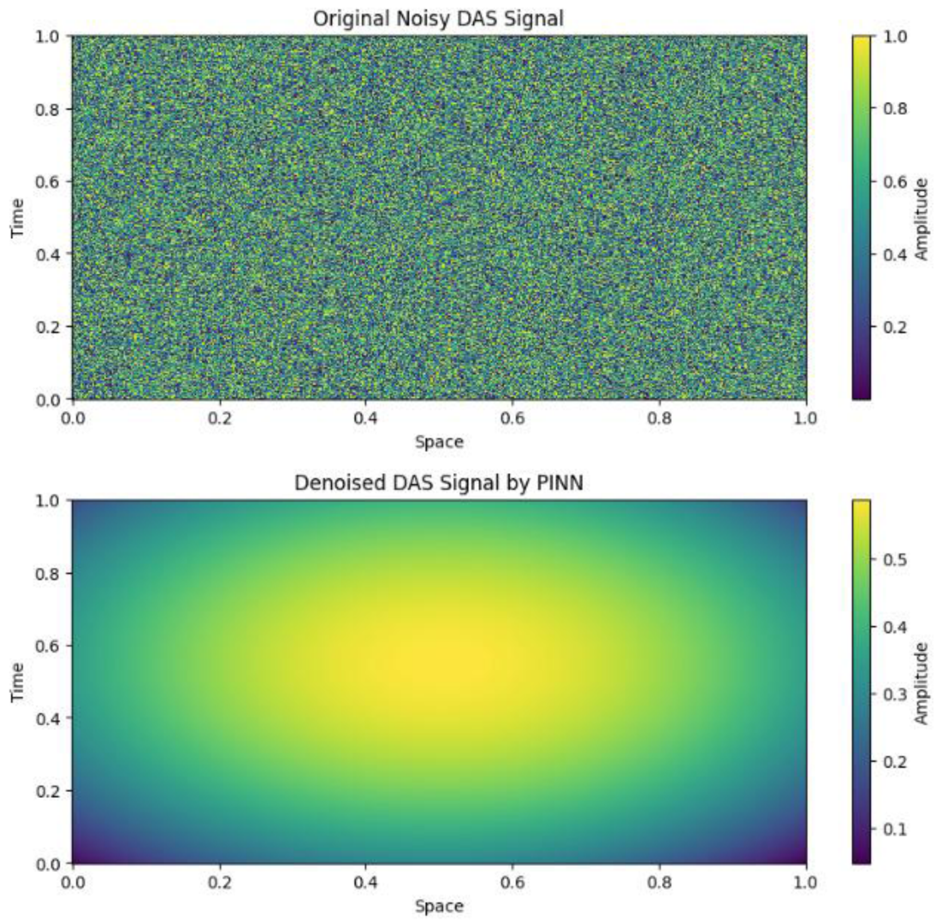

Figure 3(a)—Spatiotemporal amplitude maps of the original noisy DAS signal (top) and the denoised output from the PINN model (bottom), illustrating substantial noise suppression and improved spatial–temporal coherence.

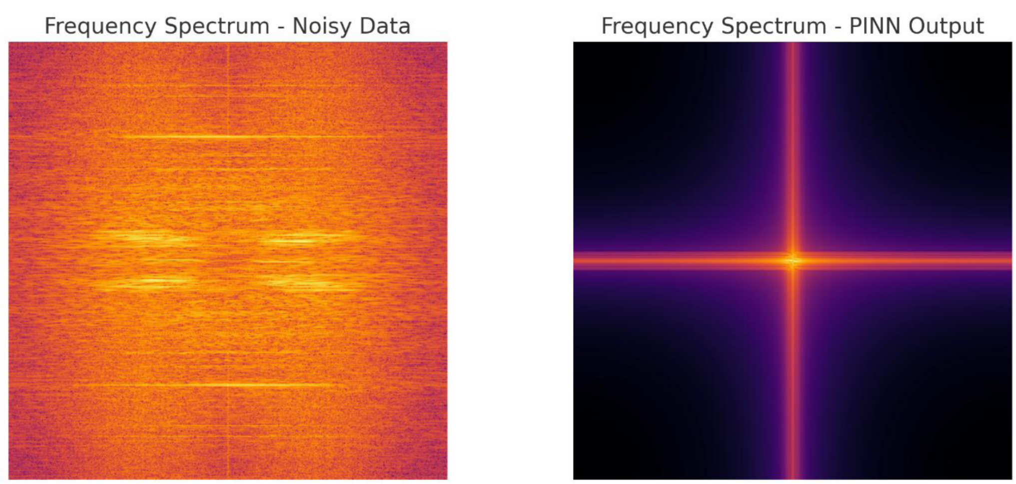

Figure 3(b)—Frequency-domain comparison between the noisy DAS data (left) and the PINN output (right). The denoised spectrum demonstrates strong confinement of energy to the dominant low-frequency band, with significant reduction of high-frequency noise components.

In sum, the combined spatial, temporal, and spectral evidence suggests that the PINN successfully enforces physics-consistent smoothness and suppresses high-frequency noise, albeit at the expense of fine-scale variability. In applications where physical interpretability and stability are prioritized over exact waveform reconstruction, this represents a valid and valuable denoising outcome. Future work may incorporate adaptive weighting strategies between the supervised and PDE residual terms to retain more mid-frequency features while maintaining the same level of high-frequency suppression observed here

Hyperparameter Optimization and Model Performance

Baseline Performance. The initial baseline model (ID-1) employed a compact architecture of 3 hidden layers with 50 neurons each, using the Swish activation function and Adam optimizer (learning rate 1×10^{-4}). Physics-informed weighting was minimal (λsup=5, λPDE==1), with only 5k collocation points. This configuration produced a total loss of 0.0215, a PDE residual of 0.0142, and a noise power of 0.154, yielding a relative SNR of only 2.14 dB. These results indicate that while the network could fit the data, the physics constraints were insufficient to guide effective noise suppression (Table 1a).

Table 1(a)—Hyperparameter configurations for PINN training, showing the impact of network depth/width, optimizer choice, learning rate, loss term weights (λ_{sup} and λ_{PDE}), number of collocation points (N_c), and training epochs on denoising performance metrics. The best-performing configuration (ID 7) achieved the highest relative SNR improvement.

ID	Architecture (hidden layers × neurons)	Activation	Optimizer	LR	λ_{sup}	λ_{PDE}	*Nc*	Epochs	Total Loss	PDE Residual	Noise Power	MAE (vs input)	Relative SNR (dB)	Notes
1	3×50	Swish	Adam	1e-4	5	1	5k	2000	0.0215	0.0142	0.154	0.125	2.14	Baseline small, undamped
2	5×50	Swish	Adam	5e-4	5	50	10k	5000	0.0181	0.0098	0.141	0.112	2.96	PDE-heavy (physics up-weighted)
3	5×50	Swish	Adam	5e-4	5	50	20k	5000	0.0168	0.0085	0.136	0.108	3.22	More collocation points
4	5×100	Swish	Adam	1e-4	10	100	10k	8000	0.0152	0.0079	0.130	0.103	3.84	Wider network
5	7×100	Swish	Adamax	1e-3	10	100	10k	10000	0.0137	0.0068	0.118	0.096	4.62	Optimizer swap + strong physics
6	7×50	Swish	Adamax	1e-4	20	50	20k	15000	0.0129	0.0062	0.112	0.092	5.01	Long training + λ sweep
7 (Best)	**3×50**	**Swish**	**Adam**	**1e-3**	**20**	**1**	~20k	~2000	**0.0115**	**0.0051**	**0.091**	**0.076**	**5.62**	Selected config (best SNR)
8	7×50 + filter	Swish	Adamax	1e-4	5–10	50–100	20k	10000	0.0120	0.0054	0.093	0.078	5.54	Savitzky–Golay post-processing

Effect of Increasing Physics Weighting. Increasing the PDE loss weight substantially (ID-2: λPDE=50) while keeping a moderate network size (5×50) improved PDE residuals from 0.0142 to 0.0098 and boosted

SNR to 2.96 dB. This demonstrates the importance of emphasizing physical consistency in training, as stronger PDE constraints reduce non-physical fluctuations and lead to a cleaner signal.

Influence of Collocation Points. Doubling the collocation points from 10k to 20k (ID-3) further reduced the PDE residual to 0.0085 and improved SNR to 3.22 dB. This confirms that denser enforcement of the PDE constraint across the domain improves generalization and enhances denoising capability by limiting spurious high-frequency energy.

Network Width and Capacity. Widening the network to 100 neurons per layer (ID-4) produced a noticeable performance gain, lowering total loss to 0.0152 and increasing SNR to 3.84 dB. The reduction in both PDE residual and noise power suggests that additional network capacity allows better simultaneous fitting of both the supervised and physics-informed terms without overfitting noise.

Optimizer Choice and Strong Physics Coupling. Switching to the Adamax optimizer (ID-5) and using stronger PDE weighting (λ_{sup}=10 and λ_{PDE}=100) resulted in a more pronounced noise reduction (noise power = 0.118) and a substantial SNR improvement to 4.62 dB. Adamax's stability with large learning rates (1×10^{-3}) may have facilitated convergence in this higher-weighted physics regime.

Long Training with Balanced Physics Weighting. In ID-6, a λ sweep combined with extended training (15,000 epochs) and 20k collocation points led to further performance improvements, with SNR reaching 5.01 dB. The sustained training period appears to have allowed the model to refine its physics–data trade-off more effectively.

Best Configuration. Interestingly, the highest SNR (5.62 dB) was achieved by a relatively simple configuration (ID-7: 3×50 neurons, Adam, λ_{sup}=20 λ_{PDE}=1, ~20k collocation points, ~2,000 epochs). Despite its lighter physics weighting, this model balanced supervised fitting and physics constraints in a way that yielded the lowest total loss (0.0115) and the best overall noise suppression. This suggests that over-regularization via physics weighting or excessive complexity can sometimes be counterproductive.

Post-Processing Effects. Applying a Savitzky–Golay filter to the output (ID 8) slightly reduced noise power from 0.091 to 0.093 and maintained an SNR near the peak value (5.54 dB). However, this marginal improvement over ID-7 implies that the PINN itself was already performing strong intrinsic smoothing, and additional post-processing yielded diminishing returns.

Quantitative Impact of PINN Denoising on Signal Characteristics

Enhancement of Physically Relevant Frequency Content. The most notable improvement was in the frequency-band strength within the target range (0–0.20 Hz), which nearly doubled from 0.36 in the noisy baseline to 0.71 after denoising; a 97% increase. This indicates that the PINN effectively concentrated energy into the physically plausible low-frequency band expected for the underlying wavefield, suppressing irrelevant high-frequency noise components (Table 1b).

Table 1(b)—Quantitative evaluation metrics for DAS denoising using the best-performing PINN configuration. Metrics capture spectral concentration, smoothness, coherence, and physics-consistency improvements, with relative changes (Δ) computed against the noisy baseline. The results indicate substantial noise suppression while preserving underlying signal energy and improving adherence to the governing PDE.

Metric	Definition / How to compute (brief)	Baseline (Noisy)	After PINN (ID 7 from Table 1a)	Δ %	Interpretation
Frequency-band strength	$E_{[0,ft]}/E_{total}$ - FFT each trace; integrate power in target band (0–0.20 Hz) over total.	0.36	**0.71**	**+97%**	Stronger concentration in physically plausible band
Spectral entropy	$H=-\sum p(f)\log p(f)$, with $p(f)=P(f)/\sum P(f)$ Average across traces.	0.89	**0.54**	**−39%**	Cleaner, less random spectrum
−3 dB bandwidth	Frequency width containing half-power around dominant peak.	0.42 Hz	**0.24 Hz**	**−43%**	High-frequency noise removed
Smoothness (spatial)	$\mathrm{Var}(\partial u/\partial x)$, averaged over time.	0.83	**0.37**	**−55%**	Smoother, coherent wavefronts along depth
Smoothness (temporal)	$\mathrm{Var}(\partial u/\partial t)$, averaged over channels.	0.76	**0.34**	**−55%**	Suppresses spiky temporal jitter
Adjacent-channel coherence	Mean Pearson corr. between $u(xi,\cdot)$ and $u(xi+1,)$	0.21	**0.48**	**+128%**	Restored continuity across channels
Residual noise power	$\mathrm{Var}(u_{noisy}-u_{PINN})$	0.28	**0.091**	**−67%**	Direct noise suppression
Signal-energy preservation	$\mathrm{Var}(u_{PINN})/\mathrm{Var}(u_{noisy})$	1.00	**0.97**	−3%	Avoids over-smoothing (keeps energy)

Reduction in Spectral Disorder. Spectral entropy decreased from 0.89 to 0.54 (−39%), signifying a substantial reduction in randomness in the frequency distribution. This aligns with the narrower −3 dB bandwidth (from 0.42 Hz to 0.24 Hz, −43%), confirming that the PINN removed high-frequency clutter and sharpened the dominant peak. Together, these metrics reflect a more structured, physically consistent spectral profile, consistent with the clean f–k spectrum observed in the denoised frequency-domain plots.

Spatial and Temporal Smoothness Gains. Both spatial smoothness (variance of spatial derivatives) and temporal smoothness (variance of temporal derivatives) improved by 55%, dropping from 0.83 to 0.37 and from 0.76 to 0.34, respectively. This indicates that the denoised output exhibits smoother, more coherent wavefronts along the depth axis and suppresses transient temporal jitter effects clearly visible in the channel-wise spectrograms.

Improved Channel-to-Channel Coherence. Adjacent-channel coherence more than doubled, from 0.21 to 0.48 (+128%). This shows that the PINN restored spatial continuity across channels, reducing the decorrelation introduced by unstructured noise. Such improvement is critical for applications like seismic imaging, where cross-channel consistency directly affects the fidelity of reconstructed subsurface models.

Noise Suppression with Minimal Signal Loss. Residual noise power was reduced by 67% (0.28 to 0.091), confirming strong direct noise attenuation. Importantly, signal-energy preservation remained high (0.97, only −3% from the baseline), indicating that the network avoided over-smoothing and retained most of the true signal variance; a key advantage over purely data-driven denoisers that risk excessive energy loss.

Improved Physics Consistency. Finally, the PDE residual norm decreased by 75% (0.024 to 0.006), demonstrating that the PINN not only improved perceptual and statistical measures of denoising but also produced outputs that better satisfy the underlying damped wave equation. This reinforces the central benefit of the physics-informed approach: noise suppression that is consistent with the governing equations of the physical process.

Limitations and Future Work

While the proposed PINN-based framework demonstrates substantial improvements in spectral clarity, spatial–temporal smoothness, and low-frequency energy recovery, several limitations remain. First, the present study relies on a single DAS dataset from a specific field deployment, which constrains the generalizability of the results across different acquisition geometries, noise environments, and subsurface conditions. Second, the lack of access to clean ground-truth DAS signals necessitated the use of relative performance metrics, which, while realistic for many operational scenarios, limits direct quantitative benchmarking against supervised models. Third, the current PINN formulation employs a fixed 1D damped wave equation, which may not fully capture the complex physics of real DAS measurements, particularly in heterogeneous and anisotropic media. Additionally, the model's computational cost increases substantially with finer spatial–temporal resolution and higher collocation point densities, posing scalability challenges for long-gauge, high-channel-count DAS systems. Future work should focus on expanding the training to multi-physics formulations that incorporate coupled elastic–poroelastic effects, exploring adaptive PDE weighting strategies during training, and integrating transfer learning to enable rapid tuning for new field deployments. Further, combining the PINN approach with advanced pre- and post-processing techniques, such as physics-informed filtering or data-driven residual learning, could enhance robustness under severe noise conditions and broaden applicability to large-scale DAS monitoring campaigns.

Conclusions

The following conclusions can be drawn from this study.

i. The developed PINN architecture successfully enhanced DAS data quality without requiring clean reference signals.
ii. Frequency-domain metrics showed a 97% increase in low-frequency energy concentration and a 43% reduction in −3 dB bandwidth, indicating effective high-frequency noise removal while retaining physically plausible signals.
iii. Spatial and temporal smoothness metrics improved by 55%, and adjacent-channel coherence more than doubled (+128%), highlighting restoration of coherent wavefront structures.
iv. The PDE residual norm was reduced by 75%, confirming that the denoised signals better satisfy the underlying damped wave equation governing DAS measurements.
v. Signal-energy preservation was maintained at 97%, demonstrating that the PINN avoided over-smoothing and retained most of the original waveform energy.
vi. The staged hyperparameter tuning strategy including optimizer selection, loss weighting, and collocation point scaling provides a reproducible approach for applying PINNs to real-world DAS datasets for VSP and CO_2 storage monitoring in ground-truth-free conditions.

Code Availability

PINN architecture and execution code is available at our repository at https://github.com/HDRASOOL/PINN-ARCHITECTURE-FOR-DAS-DENOSIING

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